

# Functions — Practice-Style 8 Questions

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Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

Time: 20–30 minutes

Calculator: Allowed

## Instructions

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- Answer all questions.
- Show full working where appropriate.
- Leave exact answers unless told to round.
- Use correct mathematical notation.
- Keep your work organized using the question numbers.

## Quick formulas

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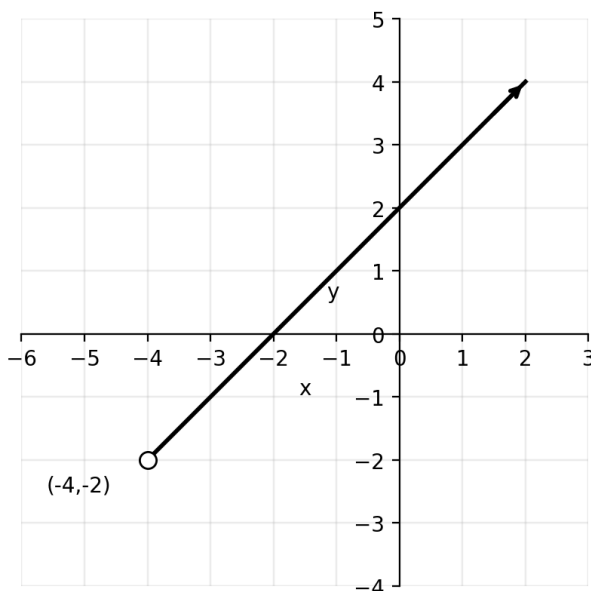
- For  $y = \sqrt{\text{expression}}$ , require  $\text{expression} \geq 0$
- For  $y = \frac{1}{\text{expression}}$ , require  $\text{expression} \neq 0$
- For  $y = \frac{1}{\sqrt{\text{expression}}}$ , require  $\text{expression} > 0$
- A function has an inverse only if it is one-to-one on its stated domain

## Questions

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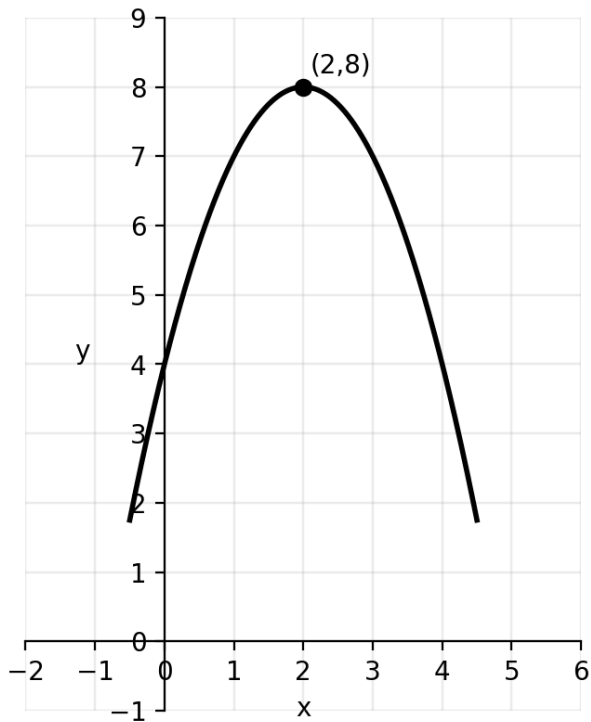
1. Write down the domain and range of each graph.

(a)



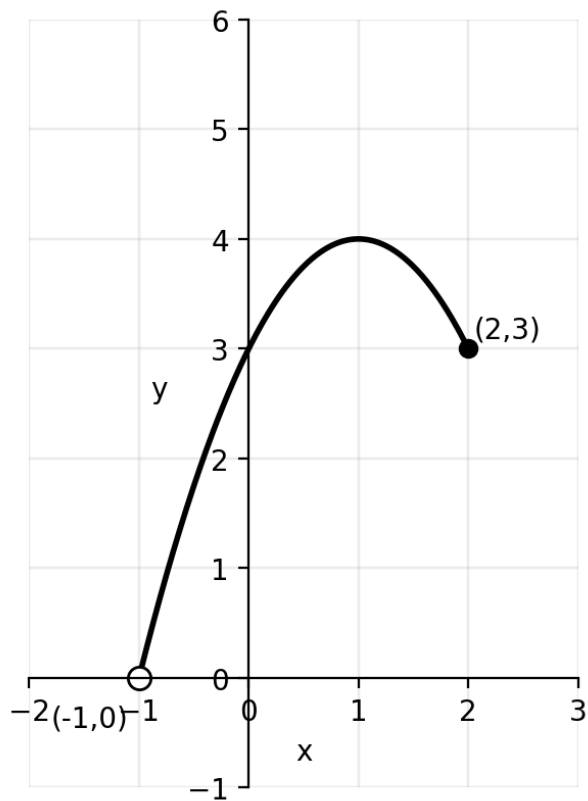
The graph is a ray starting at the open point  $(-4, -2)$  and going up to the right.

**(b)**



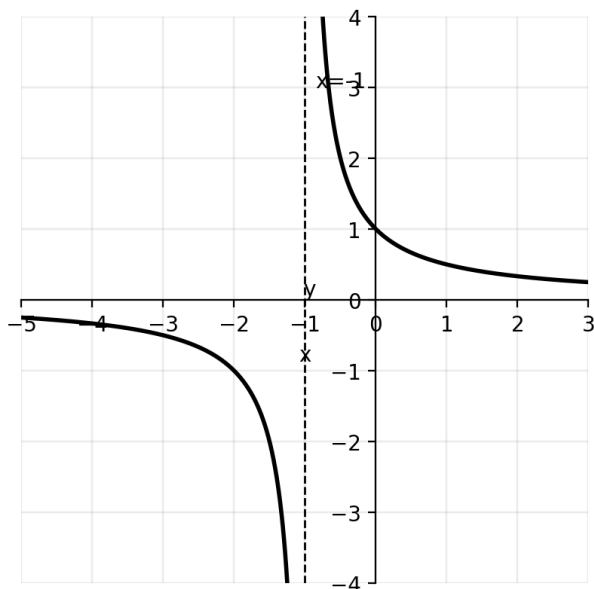
The graph is a downward-opening parabola with maximum point  $(2, 8)$ .

**(c)**



The curve begins at the open point  $(-1, 0)$  and ends at the closed point  $(2, 3)$ .

**(d)**



The graph has a vertical asymptote at  $x = -1$  and a horizontal asymptote at  $y = 0$ .

2. Find the largest possible domain of each function.

(a)  $y = \sqrt{5x - 6} + 3$

(b)  $y = \frac{4x + 7}{x + 2}$

(c)  $y = \frac{2}{\sqrt{3x - 1}}$

(d)  $y = \log_2(x + 9)$

3. You are given the function

$$f(x) = -x^4 + 6x^2 - 2$$

defined for  $0 \leq x \leq 2$ .

Write down the range of  $f$ .

4. Given  $f(x) = 3x$  and  $g(x) = 2x - 1$ , write down the values of:

(a)  $f \circ g(3)$

(b)  $g \circ f(2)$

(c)  $g \circ g(0)$

(d)  $f \circ g \circ f(-1)$

5. Given  $f(x) = 6x - 1$  and  $g(x) = \frac{x}{6} + 4$ , find in simplest form:

(a)  $f \circ g(x)$

**(b)**  $g \circ f(x)$

**(c)**  $f \circ f(x)$

**6.** Given  $f(x) = x + 5$  and  $g(x) = 4x - 12$ , find in the form  $ax + b$ :

**(a)**  $f^{-1}(x)$

**(b)**  $g^{-1}(x)$

**(c)**  $f^{-1} \circ g^{-1}(x)$

**(d)**  $g^{-1} \circ f^{-1}(x)$

**7.** Consider the functions

$$f(x) = 2x^2 - 16x + 35 \quad \text{and} \quad g(x) = x - 1$$

**(a)** Write down the largest possible domain and range of  $f(x)$ .

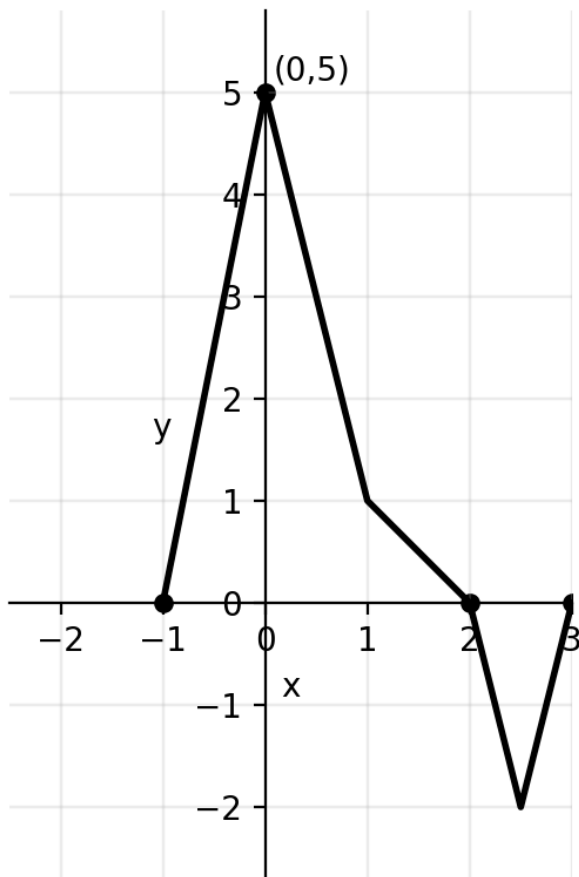
**(b)** Let  $h(x) = f \circ g(x)$ . Find an expression for  $h(x)$  in the form  $ax^2 + bx + c$ .

**(c)** Expand and simplify  $2(x - 5)^2 + 3$ .

**(d)** The domain of  $h(x)$  is now limited to  $x \geq a$  such that this function has an inverse. Write down the smallest possible value of  $a$ .

**(e)** For the value of  $a$  found in part (d), find an expression for  $h^{-1}(x)$ .

**8.** Let  $f(x) = 2x + 5$ . The graph of  $g(x)$  is shown below.



From the graph,  $g(0) = 5$ ,  $g(2) = 0$ , and  $g(-1) = 0$ .

Write down the value of:

(a)  $f \circ g(0)$

(b)  $\sqrt{g \circ f(-3)}$

## Answer Key

1.

(a)  $D : x > -4, R : y > -2$

(b)  $D : x \in \mathbb{R}, R : y \leq 8$

(c)  $D : -1 < x \leq 2, R : 0 < y \leq 3$

(d)  $D : x \neq -1, R : y \neq 0$

2.

(a)

$$5x - 6 \geq 0 \Rightarrow x \geq \frac{6}{5}$$

(b)

$$x + 2 \neq 0 \Rightarrow x \neq -2$$

(c)

$$3x - 1 > 0 \Rightarrow x > \frac{1}{3}$$

(d)

$$x + 9 > 0 \Rightarrow x > -9$$

3.

$$f(0) = -2, \quad f(2) = 6$$

$$f'(x) = -4x^3 + 12x = -4x(x^2 - 3)$$

Critical value in the interval:  $x = \sqrt{3}$

$$f(\sqrt{3}) = 7$$

So the range is

$$-2 \leq f(x) \leq 7$$

4.

(a)  $g(3) = 5$ , so  $f(g(3)) = f(5) = 15$

(b)  $f(2) = 6$ , so  $g(f(2)) = g(6) = 11$

(c)  $g(0) = -1$ , then  $g(-1) = -3$

(d)  $f(-1) = -3$ , then  $g(-3) = -7$ , then  $f(-7) = -21$

5.

(a)

$$f \circ g(x) = 6 \left( \frac{x}{6} + 4 \right) - 1 = x + 23$$

(b)

$$g \circ f(x) = \frac{6x - 1}{6} + 4 = x + \frac{23}{6}$$

(c)

$$f \circ f(x) = 6(6x - 1) - 1 = 36x - 7$$

6.

(a)  $f^{-1}(x) = x - 5$

(b)  $g^{-1}(x) = \frac{x + 12}{4} = \frac{x}{4} + 3$

(c)

$$f^{-1} \circ g^{-1}(x) = \frac{x}{4} - 2$$

(d)

$$g^{-1} \circ f^{-1}(x) = \frac{x+7}{4}$$

7.

(a)

$$f(x) = 2x^2 - 16x + 35 = 2(x-4)^2 + 3$$

Largest domain:  $x \in \mathbb{R}$

$$R : f(x) \geq 3$$

(b)

$$h(x) = f(x-1) = 2(x-1)^2 - 16(x-1) + 35 = 2x^2 - 20x + 53$$

(c)

$$2(x-5)^2 + 3 = 2x^2 - 20x + 53$$

(d)

$$h(x) = 2x^2 - 20x + 53 = 2(x-5)^2 + 3$$

So  $a = 5$ .

(e)

$$y = 2(x-5)^2 + 3$$

$$y-3 = 2(x-5)^2$$

Since  $x \geq 5$ ,

$$x = 5 + \sqrt{\frac{y-3}{2}}$$

Therefore

$$h^{-1}(x) = 5 + \sqrt{\frac{x-3}{2}}$$

8.

(a)

$$f \circ g(0) = f(5) = 15$$

(b)

$$f(-3) = -1, \quad g(-1) = 0, \quad \sqrt{g \circ f(-3)} = 0$$